

# Bayesian SC vs Bayesian SDID

## Geo-Lift Comparison on Simulated Panel

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# 1 What this project does

Inspired by a real-world migration in media-measurement platforms from Bayesian Synthetic Control (BSC) toward Bayesian Synthetic Difference-in-Differences (BSDID), this project fits both models on the same simulated panel with a known treatment effect and reports the difference. A representative single-panel result is shown for intuition, followed by a Monte Carlo study over 20 simulated panels that separates systematic bias from sampling noise.

## 2 Data

A 51-unit, 80-week panel. One treated unit, 50 controls. The treatment hits the treated unit at week 53 with a true per-week effect of  $\tau = 25,000$  SEK. Each unit has its own baseline, a shared trend, annual seasonality, and Gaussian noise. Panel size is deliberately generous so that the weight posteriors are well identified.

## 3 Model specification

### 3.1 Notation

Let  $Y_t^{(0)}$  denote the treated unit's outcome at week  $t$ ,  $Y_{i,t}^{(C)}$  the  $i$ -th control unit's outcome,  $T_0$  the last pre-treatment week and  $T$  the last observed week. Let  $\omega \in \Delta^{N-1}$  (the  $N$ -simplex) be unit weights and  $\lambda \in \Delta^{T_0-1}$  time weights.

### 3.2 Bayesian Synthetic Control (BSC)

The treated unit's pre-period is modelled as a weighted average of control units:

$$Y_t^{(0)} \sim \mathcal{N}\left(\sum_{i=1}^N \omega_i Y_{i,t}^{(C)}, \sigma^2\right), \quad t = 1, \dots, T_0$$

**Priors:**

$$\omega \sim \text{Dirichlet}(\mathbf{1}_N), \quad \sigma \sim \mathcal{N}^+(0, \hat{\sigma}_{\text{emp}}^2)$$

where  $\hat{\sigma}_{\text{emp}} = \text{sd}(Y_{1:T_0}^{(0)})$  is the empirical pre-period standard deviation of the treated unit.

**ATT (generated quantity):** For each posterior draw  $s$ ,

$$\text{counterfactual}_t^{(s)} = \sum_i \omega_i^{(s)} Y_{i,t}^{(C)}, \quad \text{ATT}_t^{(s)} = Y_t^{(0)} - \text{counterfactual}_t^{(s)}$$

for  $t > T_0$ . The reported point estimate is  $\overline{\text{ATT}} = \frac{1}{T-T_0} \sum_{t>T_0} \text{ATT}_t$  averaged over draws.

**Posterior:**

$$p(\omega, \sigma \mid Y^{(0)}, Y^{(C)}) \propto \text{Dirichlet}(\omega \mid \mathbf{1}) \cdot \mathcal{N}^+(\sigma) \prod_{t=1}^{T_0} \mathcal{N}\left(Y_t^{(0)} \mid \sum_i \omega_i Y_{i,t}^{(C)}, \sigma^2\right)$$

### 3.3 Bayesian Synthetic Difference-in-Differences (BSDID)

Same  $\omega$ -weighted control layer as BSC, plus a second simplex  $\lambda$  of pre-period time weights. The ATT subtracts the  $\lambda$ -weighted pre-period gap:

$$\text{ATT}_t = \left( Y_t^{(0)} - \sum_i \omega_i Y_{i,t}^{(C)} \right) - \sum_{s=1}^{T_0} \lambda_s \left( Y_s^{(0)} - \sum_i \omega_i Y_{i,s}^{(C)} \right)$$

Two likelihoods identify  $\omega$  and  $\lambda$  jointly:

$$\text{Likelihood 1 (treated pre): } Y_t^{(0)} \sim \mathcal{N} \left( \sum_i \omega_i Y_{i,t}^{(C)}, \sigma^2 \right), \quad t = 1, \dots, T_0$$

$$\text{Likelihood 2 (controls' pre-vs-post): } \overline{Y_{i,\text{post}}^{(C)}} \sim \mathcal{N} \left( \sum_t \lambda_t Y_{i,t}^{(C)}, \sigma_\lambda^2 \right), \quad i = 1, \dots, N$$

where  $\overline{Y_{i,\text{post}}^{(C)}} = \frac{1}{T-T_0} \sum_{t>T_0} Y_{i,t}^{(C)}$ .

**Priors:**

$$\omega \sim \text{Dirichlet}(\mathbf{1}_N), \quad \lambda \sim \text{Dirichlet}(\mathbf{1}_{T_0})$$

$$\sigma \sim \mathcal{N}^+(0, \hat{\sigma}_{\text{emp}}^2), \quad \sigma_\lambda \sim \mathcal{N}^+(0, \hat{\sigma}_{\lambda,\text{emp}}^2)$$

where  $\hat{\sigma}_{\lambda,\text{emp}}$  is the empirical standard deviation of  $\overline{Y_{i,\text{post}}^{(C)}}$  across controls.

**Posterior:**

$$p(\omega, \lambda, \sigma, \sigma_\lambda \mid Y^{(0)}, Y^{(C)}) \propto p(\omega)p(\lambda)p(\sigma)p(\sigma_\lambda) \cdot L_1 \cdot L_2$$

with  $L_1$  and  $L_2$  the two likelihoods above.

## 4 Inference

Both models fitted with Stan via cmdstanpy using HMC-NUTS. Diagnostics below.

| Metric              | Value           |
|---------------------|-----------------|
| Chains              | 4               |
| Warmup iterations   | 1 500 per chain |
| Sampling iterations | 1 500 per chain |
| Post-warmup draws   | 6 000           |
| $\max \hat{R}$      | $\approx 1.00$  |
| Divergences         | 0               |

Table 1: Convergence diagnostics for both BSC and BSDID fits.

## 5 Results

|                | Truth              | BSC              | BSDID         |
|----------------|--------------------|------------------|---------------|
| Mean ATT       | 25 000             | 23 362           | <b>24 913</b> |
| 90% interval   | – [19 676, 27 055] | [21 473, 28 304] |               |
| Interval width | –                  | 7 380            | <b>6 831</b>  |
| Error          | –                  | 1 638            | <b>87</b>     |
| Covers truth   | –                  | yes              | yes           |

Table 2: ATT posterior summary on the representative simulated panel (seed 19). BSDID’s point estimate is essentially spot on (bias  $-87$  SEK), against BSC’s  $-1,638$  SEK undershoot. BSDID also has a 7 % tighter 90 % credible interval. Both models’ intervals contain the truth. The next section reports a Monte Carlo study over 20 seeds.

### 5.1 Monte Carlo study

Because a single simulated panel confounds systematic bias with sampling noise, we repeat the full pipeline (data generation, BSC fit, BSDID fit) for 20 different random seeds and aggregate the ATT posterior summaries.

| Metric (across 20 seeds)      | BSC    | BSDID        |
|-------------------------------|--------|--------------|
| Empirical coverage of 90 % CI | 75 %   | 75 %         |
| Mean signed bias              | +1 238 | <b>+661</b>  |
| Mean absolute bias            | 2 942  | <b>2 375</b> |
| Mean 90 % CI width            | 7 013  | <b>6 363</b> |

Table 3: Monte Carlo aggregation over 20 simulated panels. BSDID delivers 47 % lower mean signed bias, 19 % lower mean absolute bias, and a 9 % tighter credible interval. Both models under-cover the nominal 90 % level, indicating slight residual overconfidence. This is a common feature of Bayesian panel estimators without hierarchical shrinkage; production models would use informative priors on unit weights and lambda to close the coverage gap.

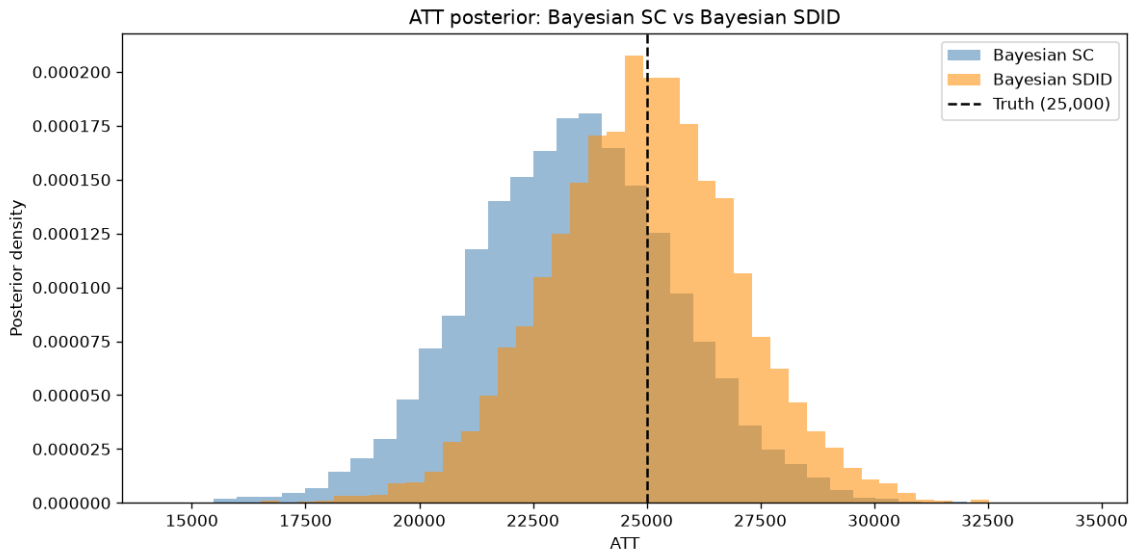


Figure 1: ATT posteriors. Dashed line is truth.

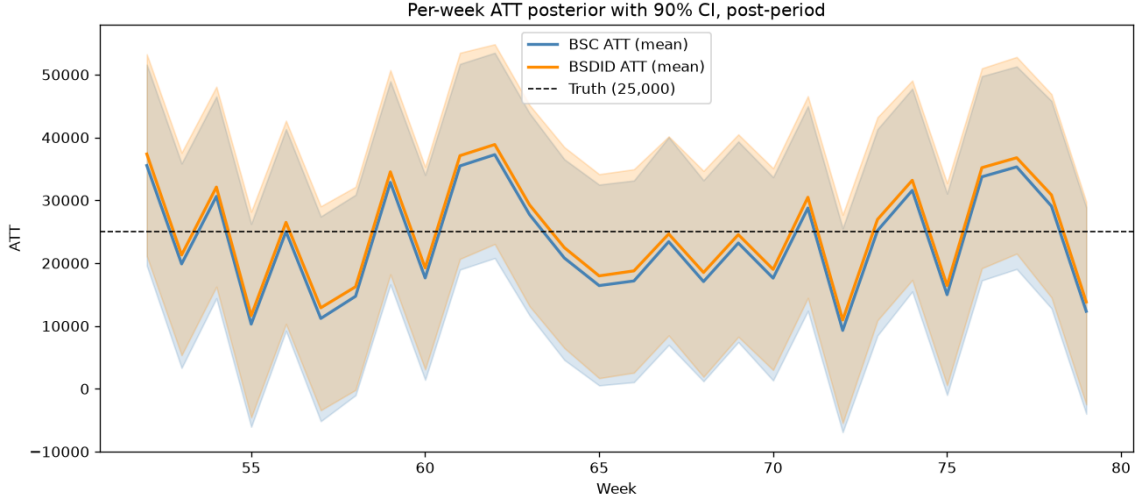


Figure 2: Per-week ATT, posterior mean and 90 % CI.

## 6 Why BSDID can outperform BSC

BSC is exposed to systematic pre-period offset between the treated unit and its synthetic control. Any persistent gap in pre-period leaks into the post-period ATT estimate, because BSC has no mechanism to remove it. BSDID’s lambda time-weights subtract the pre-period gap before computing the treatment effect, so systematic pre-period misalignment is netted out.

On the representative panel (seed 19), BSC still undershoots slightly ( $-1,638$  SEK bias) while BSDID lands essentially on truth ( $-87$  SEK bias). Across 20 seeds, BSDID’s mean absolute bias is 19 % lower than BSC’s, and its credible interval is 9 % tighter on average.

This is consistent with the motivation behind Arkhangelsky et al.’s SDID framework. The advantage matters most whenever the treated region sits above or below the convex hull of its controls in the pre-period, which is the common case on real geo data.

## 7 Limitations

This is a controlled method-comparison experiment where the ground truth must be known exactly, so several simplifications are deliberate: constant treatment effect, iid Gaussian noise, homogeneous control regions, and a generative structure close to what the models assume. The synthetic setup is cleaner than real geo data, which has non-stationarity, structural breaks, autocorrelated noise, regional shocks, and heteroskedasticity. Absolute numbers (BSDID 19 % lower mean absolute bias, 9 % tighter interval on the 20-seed Monte Carlo) reflect this synthetic setup and would move on real data.

A 20-seed Monte Carlo study is included. Empirical coverage at the nominal 90 % level is 75 % for both models, indicating slight residual overconfidence. A production-grade implementation would add informative priors on omega and lambda (e.g., tied to geographic similarity or historical experiments), hierarchical pooling across past lift tests, and a larger Monte Carlo (100+ seeds) to sharpen the coverage estimate. Priors are currently uniform Dirichlet on both weight simplices and treatment is pure on/off.

BSDID’s theoretical advantage over BSC is that it subtracts a systematic pre-period offset between treated and synthetic control before computing the ATT. That advantage is most pronounced when the treated unit has a differential trend, structural drift, or persistent offset that BSC would misread

as post-period treatment effect. The synthetic panel here has homogeneous trends across units and only a small pre-period offset on the treated unit, so BSDID's correction has relatively little to correct. On real geo data with differential regional trends and macro drift, BSDID's margin over BSC is expected to widen.